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Thermal module and joining method for hermetically sealed enclosure of a thermal module using a capillary joint

Abstract

A thermal module includes parts with diverging or divergent walls that cause a capillary pressure gradient in a bonding paste used to secure the parts together. Based in part on the capillary pressure, the bonding paste is drawn in between the divergent walls, including small gaps between the walls. Additionally, during a reflow operation of the bonding paste, particles and/or air bubbles can egress from the bonding paste, allowing the bonding paste to freeze and form a homogeneous structure. By removing particles and air bubbles, the bonding paste can form a more robust structural joint and a hermetic seal for the thermal module. The multiple parts of the thermal module can be formed into their respective shapes prior to be joined together by the bonding paste. As a result, the formation of the thermal module is less likely to damage other structures, such as wick structures.

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Background/Summary

TECHNICAL FIELD

(1) This application is directed to thermal modules, and more particularly, to thermal modules with parts having divergent walls that cause capillary forces in a bonding paste to position it between the divergent walls while it is in a molten state.

BACKGROUND

(2) Thermal modules (e.g., heat pipe, vapor chamber) are used to transport heat (e.g., thermal energy) generated by a heat-generating component. When used in electronic devices, thermal modules can transport heat away from a heat-generating component (e.g., integrated circuit), thus

allowing the heat-generating component to cool more rapidly. One solution to forming a thermal module includes bending or deforming a metal part from a circular cross section to a desired cross sectional shape, such as a lozenge shape. However, the formation process can damage one or more wick structures positioned along an inner surface of the metal, resulting in compromised performance of the thermal module. Another solution includes modifying a multi-piece thermal module to include flanges, and joining the pieces together at their respective flanges using diffusion bonding. However, the addition of flanges results in a smaller thermal module with less thermal transport performance. Alternatively, a thermal module built with a larger size/volume may be further increased due to the flanges, causing the thermal module to occupy additional space within an electronic device, which may lead to less space in the electronic device for other components.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Certain features of the subject technology are set forth in the appended claims. However, for purpose of explanation, several embodiments of the subject technology are set forth in the following figures.
- (2) FIG. 1 illustrates an example of an electronic device, in accordance with aspects of the present disclosure.
- (3) FIG. 2 illustrates a perspective view of an example of a thermal module, in accordance with aspect of the present disclosure.
- (4) FIG. 3 illustrates a cross sectional view of a thermal module, in accordance with aspect of the present disclosure.
- (5) FIGS. 4A, 4B, and 4C illustrate a process for joining two housing parts of a thermal module together, in accordance with aspect of the present disclosure.
- (6) FIG. 5 illustrates a cross sectional view of a thermal module, showing the walls of the housing parts diverging at a different angle, in accordance with aspect of the present disclosure.
- (7) FIG. 6 illustrates a perspective view of a housing part of a thermal module modified to include several openings along a surface of a wall, in accordance with aspect of the present disclosure.
- (8) FIG. 7 illustrates an enlarged perspective partial cross sectional view of housing parts of a thermal module joined together, with one of the walls having openings, in accordance with aspect of the present disclosure.
- (9) FIGS. 8-10 illustrate cross sectional views of alternate examples of a thermal module, in accordance with aspect of the present disclosure.
- (10) FIGS. 11 and 12 illustrate an alternate example of a thermal module, showing a port used to receive a refrigerant, in accordance with aspect of the present disclosure.
- (11) FIGS. 13 and 14 illustrate alternate examples of thermal modules having different shapes, in accordance with aspect of the present disclosure.
- (12) FIGS. 15A and 15B illustrate alternate examples of electronic devices that may include one or more thermal modules shown and described herein.
- (13) FIG. 16 illustrates a flowchart showing a process for assembling a thermal module, in accordance with aspect of the present disclosure.
- (14) FIG. 17 illustrates a flowchart showing a process for joining housing parts that form an enclosure, in accordance with aspect of the present disclosure.

DETAILED DESCRIPTION

- (15) The detailed description set forth below is intended as a description of various configurations of the subject technology and is not intended to represent the only configurations in which the subject technology may be practiced. The appended drawings are incorporated herein and constitute a part of the detailed description. The detailed description includes specific details for the

purpose of providing a thorough understanding of the subject technology. However, it will be clear and apparent to those skilled in the art that the subject technology is not limited to the specific details set forth herein and may be practiced without these specific details. In some instances, well-known structures and components are shown in block diagram form in order to avoid obscuring the concepts of the subject technology.

(16) The subject technology is directed to thermal modules with multiple parts (e.g., housing parts) secured together, with respective walls of the parts modified to promote flow of a bonding paste between the walls. For example, one of the walls may be formed at an angle (e.g., non-zero) such that the walls are divergent from, and non-parallel with respect to, each other. Accordingly, respective walls of the parts may extend in different directions. The bonding paste may be selected from copper paste or solder paste, as non-limiting examples. Accordingly, at room temperature the bonding paste is in a solid state, or solid form, and is converted to a liquid state, or liquid form, when heated to a temperature at or above the melting temperature of the bonding paste.

Beneficially, the bonding paste in liquid state incurs a force (e.g., capillary pressure) and is drawn, or forced, between a gap of the divergent walls, thus increasing the contact surface area between the bonding paste and the walls. The bonding paste in a solid state thereafter provides a structural joint and hermetic seal between the parts of the thermal module. Capillary forces cause the molten liquid to form a concave meniscus due to the divergence angle between the opposing walls, and the capillary pressure gradient draws the molten liquid toward the root of the divergence and, in combination with buoyancy forces, purges bubbles of lower-density fluids (e.g., air, flux) out of the molten body through the meniscus surface. Once the molten liquid freezes back into a solid state, the joint is formed of a non-porous solid that provides a hermetic seal between the two now-joined parts of the enclosure.

(17) The angle between the divergent walls can influence the force at which the bonding paste is drawn between the walls. In this regard, the angle may be selected so as to provide sufficient force to draw the bonding paste between the divergent, but limited so as to not force the bonding paste into an interior space (between the parts) and make contact with a wick structure(s) located in an interior volume, or space, between the parts.

(18) Additionally, while heated and maintained in its liquid state, particles (e.g., lower-density fluids or particles, solder flux) and/or air bubbles can be forced out of the bonding paste due in part to the difference between the respective densities of the molten bonding paste and the particles, fluids, or air bubbles. When the bonding paste is removed from heat and reaches its freezing point, the bonding paste may form a homogeneous structure free, or substantially free, of particles and air bubbles. Beneficially, the structural joint and hermetic seal provided by the bonding paste is enhanced by the removal of the particles, fluids, and air bubbles.

(19) These and other embodiments are discussed below with reference to FIGS. 1-17. However, those skilled in the art will readily appreciate that the detailed description given herein with respect to these Figures is for explanatory purposes only and should not be construed as limiting.

(20) According to some embodiments, for example as shown in FIG. 1, an electronic device **100** takes the form of a laptop computing device. Electronic device **100** may include a housing component **102a** (e.g., display housing) and a housing component **102b** (e.g., base). The housing components **102a** and **102b** may be rotationally coupled together, thus allowing relative rotational movement between housing components **102a** and **102b**. As shown, housing component **102a** carries a display **104** designed to present visual information (e.g., textual information, still images, video images, or a combination thereof), while housing component **102b** carries an input mechanism **106a** (e.g., track pad) and an input mechanism **106b** (e.g., keyboard).

(21) Additionally, electronic device **100** may include processing circuitry connected (e.g., electrically connected) to display **104** and input mechanisms **106a** and **106b**. In this regard, electronic device **100** may include a heat-generating component **108** that takes the form of processing circuitry, which may include a central processing unit, a graphics processing unit, one or

more microcontrollers, one or more micro electromechanical system (MEMS) controllers, an application-specific integrated circuit, or a combination thereof. Electronic device **100** may further include a circuit board **110** on which heat-generating component **108** is located. In order to draw heat away from heat-generating component **108**, electronic device **100** may further include a thermal module **120** (representing one or more thermal modules) that is thermally coupled to heat-generating component **108**. In some examples, thermal module **120** is a heat pipe. Alternatively, in some examples, thermal module **120** is a vapor chamber. Electronic device **100** may include a combination of a heat pipe(s) and a vapor chamber(s). Thermal module **120** is designed to transport heat generated by heat-generating component **108** to another location within electronic device **100**, where the transported heat can be dissipated and/or expelled from electronic device **100**. For example, electronic device **100** may include a fin stack (not shown in FIG. 1) thermally coupled to thermal module **120**, with the fin stack designed to remove heat from thermal module **120**. Also, although not expressly shown, electronic device **100** may include one or more additional thermal modules with similar or different shapes and sizes, as compared to thermal module **120**.

(22) Referring to FIG. 2, a thermal module **220** is shown. Thermal module **220** may include multiple parts (e.g., multiple housing parts) that couple together. Also, thermal module **220** may include an end **222a** and an end **222b** opposite end **222a**. When thermal module **220** is thermally coupled and placed within proximity to a heat-generating component (not shown in FIG. 2) at end **222a**, the heat received at end **222a** can be transported (in a general direction of an arrow **223**) to end **222b**, where the heat is removed from thermal module **220** at or near end **222b**. In this regard, end **222a** and end **222b** may be characterized as an evaporator region and a condenser region, respectively.

(23) Referring to FIG. 3, a cross sectional view of thermal module **220** is shown. Thermal module **220** may include a part **224a** and a part **224b** coupled to part **224a**. Based on their relative positions, part **224a** and part **224b** may be referred to as a top (or upper) part and a bottom (or lower) part, respectively. Each of parts **224a** and **224b** may include a metal (or metals) having a relatively high thermal conductivity. As non-limiting examples, each of parts **224a** and **224b** may include copper or a copper alloy. Also, each of parts **224a** and **224b** may include one or more walls. For example, part **224a** and part **224b** include a wall **226a** and a wall **226b**, respectively. Each of walls **226a** and **226b** may represent a side wall extending from a major wall, with the major wall representing a wall of greatest area of the parts **224a** and **224b**. Further, based on their relative positions, wall **226a** and wall **226b** may be referred to as an exterior (or external) wall and an interior (or internal) wall, respectively, as wall **226a** is external to and generally covers wall **226b**.

(24) Thermal modules described herein may include one or more wick structures. For example, thermal module **220** may include a wick structure **228**. In some examples, wick structure **228** includes a sintered, porous structure. Subsequent to a refrigerant **230** (e.g., water), in gas (or vapor) form, carrying thermal energy to a condenser region (not shown in FIG. 3), wick structure **228** designed to transport, using capillary forces, refrigerant **230**, in a liquid state, from the condenser region to an evaporator region.

(25) As shown in the enlarged view, walls **226a** and **226b** may be secured together by a bonding paste **232**, or bonding material. In some examples, bonding paste **232** includes a solder paste, a copper paste, or a brazing paste. Other materials or substances (e.g., adhesives) may be selected. Thermally cured adhesives can behave in a similar way as a solder or brazing paste. For example, an epoxy adhesive that is cured with a baking process will undergo an initial decrease in viscosity as its temperature is elevated, at which point it can easily wick into the root of the divergent gap between the walls and form a meniscus there. After a time, the epoxy will set and become a solid body in the gap, thereby forming a mechanical joint and a hermetic seal in a controlled location between the two parts. Such adhesives may not be suitable for a thermal module, as their glass transition temperature will typically be in a range that will cause them to soften when subjected to heat produced at the evaporator zone. However, adhesive joints in non-thermal applications can be

used as conjunction with the divergent wall concept described herein to hermetically join parts. By securing walls **226a** and **226b** together, bonding paste **232** may secure parts **224a** and **224b** together. Moreover, bonding paste **232** may extend along, and fill, a gap **233** (e.g., location) between walls **226a** and **226b**. In this regard, bonding paste **232** may provide a joint (e.g., structural joint) between walls **226a** and **226b**. Moreover, bonding paste **232** may provide a hermetic seal (e.g., where gap **233** is filled by bonding paste **232**), thus keeping air and liquids out of thermal module **220**, and keeping refrigerant **230** within thermal module **220**.

(26) Also, the walls **226a** and **226b** may be characterized as divergent walls. For example, while wall **226a** is generally a vertical wall (extending along a Z-axis in Cartesian coordinates) forming a 90-degree angle (e.g., right angle) with another wall of part **224a**, wall **226b** is formed an angle **234** with respect to wall **226a**. Accordingly, the respective directions represented by the position of walls **226a** and **226b** may intersect. Put another way, walls **226a** and **226b** are non-parallel with respect to each other. Angle **234** may be approximately in the range of 10-30 degrees. Accordingly, angle **234** is a non-zero angle. As shown in FIG. 3, angle **234** is 20 degrees (or approximately 20 degrees). The divergent relationship between may walls **226a** and **226b** may exist each location of walls **226a** and **226b**, thus providing the divergence around a perimeter of thermal module **220**. Also, the distance between walls **226a** and **226b** may be continually increasing from the “root” of the divergence, or location in which walls **226a** and **226b** are the closest (or approximately the closest) to each other, to the exterior of the enclosure defined by the parts **224a** and **224b**. Also, walls **226a** and **226b** may be angled so long as walls **226a** and **226b** have different angles, thus causing a relative divergence.

(27) Using multiple parts (e.g., parts **224a** and **224b**) for a thermal module (e.g., thermal module **220**) may provide several advantages. For example, each of parts **224a** and **224b** can be formed (e.g., stamped, bent, or otherwise modified) into their respective final shapes prior to being bonded together by bonding paste **232**. Wick structure **228** can be added to part **224a** and/or part **224b**, respectively, after the forming operations of parts **224a** and **224b**. Beneficially, and unlike traditional thermal modules, wick structure **228** is not subject to external loads or other stresses caused during the formation operations of parts **224a** and **224b**, and are less likely to become damaged, thus allowing thermal module **220** to provide its intended heat transportation functions.

(28) FIGS. 4A, 4B, and 4C show a process for providing and controlling a bonding paste **332** between respective walls of two parts of a thermal module **320**. As shown, during the application of bonding paste **332**, thermal module **320** may be rotated 180 degrees (i.e., turned upside down) to maintain bonding paste **332** between a wall **326a** and a wall **326b** of a part **324a** and a part **324b**, respectively, of thermal module **320**. Also, thermal module **320** may include any features and/or characteristics for a thermal module shown and described herein.

(29) Referring to FIG. 4A, thermal module **320** includes a part **324a** and a part **324b**. Thermal module **320** may form an internal volume **335**, representing an interior space or void. Part **324a** may be characterized as defining internal volume **335**, and part **324b** may be positioned, or disposed in internal volume **335** as shown in FIG. 4A. Also, internal volume **335** may provide a location for a wick structure **328** and a refrigerant **330**. Further, as shown, bonding paste **332** is in a solid state (e.g., non-melted form as a bond wire) and is positioned between a wall **326a** (of part **324b**) and a wall **326b** (of part **324b**). As shown in the enlarged view, bonding paste **332** may include some particles, air, or other impurities. For example, bonding paste **332** may include droplets **336** (representative of additional impurity particles or droplets), each of which may take the form of soldering flux. Additionally, bonding paste **332** may include air bubbles **338** (representative of additional bubbles). Also, in order to maintain parts **324a** and **324b** together during assembly and to limit or prevent bonding paste **332**, during reflow (e.g., in an oven), from entering internal volume **335**, a clamp **339a** and a clamp **339b** (or a weight fixture) may apply a force that holds parts **324a** and **324b**, respectively, together. When bonding paste **332** takes the form of copper paste, the copper paste may include copper particles that remain solid through the

melting process. The copper particles are not intended to be purged out, based in part on their relatively high density.

(30) Referring to FIG. 4B, thermal module 320 is placed in a heating apparatus (e.g., an oven or brazing furnace) and is heated (denoted by lines 341a and lines 341b), causing reflow of bonding paste 332. As shown, walls 326a and 326b are diverging walls that are separated by an angle 334, which may be approximately in the range of 10-30 degrees. As shown in FIG. 4, angle 334 is 20 degrees, or approximately 20 degrees.

(31) Bonding paste 332, in a molten (e.g., liquid) state, may move between walls 326a and 326b based on gravitational forces. Further, based on walls 326a and 326b forming diverging walls, capillary attraction forces in the form of capillary pressure can further draw bonding paste 332 between walls 326a and 326b, causing a meniscus 340 (e.g., concave meniscus) in bonding paste 332. The capillary pressure, P.sub.c, within bonding paste 332 is governed by

$$(32) P_c = \frac{2\sigma\cos\theta}{r} \quad \text{Eq. 1}$$

where σ is the interfacial (or surface) tension between bonding paste 332 in molten state and air (e.g., ambient air) in contact with bonding paste 332, θ is the contact angle of bonding paste 332 in molten state, and r is the radius of curvature of the molten meniscus surface. As shown in FIG. 4B, a gap 344a (e.g., distance or separation) between walls 326a and 326b along the X-direction reduces in the negative Z-direction. Accordingly, based on Eq. 1, the capillary pressure P.sub.c and surface tension in the meniscus surface create a capillary rise effect in the positive Z direction, resulting in a pressure gradient within bonding paste 332 in molten state. The forces acting on or in bonding paste 332 in molten state cause bonding paste 332 to settle between walls 326a and 326b, even as the gap or space between walls 326a and 326b become smaller. Further, a gap 344b (e.g., distance or separation) between an end surface 342 (root of the divergence) of wall 326b and a wall 346 (e.g., major wall) of part 324a represents additional divergence, as gap 344b along the Z-direction reduces in the negative X-direction. The additional divergence may further increase capillary pressure along the negative X-direction. Beneficially, bonding paste 332, in molten state, is further drawn into spaces between parts 324a and 324b, thereby causing bonding paste 332 to attach to multiple different surfaces and structures of parts 324a and 324b, thus increasing the joint provided by bonding paste 332 to parts 324a and 324b. Additionally, a gap 344c, representing an additional gap between walls 326a and 326c, may include a different dimension than that of gap 344a. As shown, gap 344b is less than gap 344a (in terms of a one-dimensional distance), and gap 344c is equal to or larger than gap 344b such that, collectively, the gap is continuously growing from the closest point between parts 224a and 224b (including their respective walls) to the location of the meniscus. In this regard, the divergence described between parts 224a and 224b (in particular, their respective walls) may include an increasing gap in the positive X-direction and further increasing in the positive Z-direction. Also, bonding paste 332 may provide a capillary seal that isolates pressure within internal volume 335 from ambient air pressure surrounding thermal module 320.

(33) Additionally, in order to improve bonding and sealing capabilities of bonding paste 332, droplets 336 (e.g., soldering or brazing flux), air bubbles 338, and other impurities should be removed from bonding paste 332 such that bonding paste 332 provides a homogeneous structure. In this regard, egress of droplets 336 and air bubbles 338 may occur while bonding paste 332 is in its molten/liquid state. The velocity v of a substance rising within bonding paste 332 (in molten state) is governed by Stoke's Law:

$$(34) v = \frac{2(\rho_p - \rho_f)gr^2}{9\eta} \quad \text{Eq. 2}$$

where $\rho_{\text{sub.p}}$ is the density of an particle or impurity (e.g., droplets 336 in liquid form or air bubbles 338) in bonding paste 332, $\rho_{\text{sub.f}}$ the density of the fluid (e.g. bonding paste 332 in molten state), g is gravitational force, r is the spherical radius of the particle or impurity (e.g.,

droplets **336** or air bubble **338**), and μ is the dynamic viscosity of bonding paste **332** in molten state. The dynamic viscosity μ of bonding paste **332** is known to decrease with increasing temperature. Thus, according to Eq. 2, the velocity v of droplets **336** or air bubbles **338** within bonding paste **332** increases with increasing temperature of bonding paste **332** in molten state. Additionally, the absolute value of the velocity, v , increases as the difference between density $\rho_{\text{sub.f}}$ of bonding paste **332** in molten state increases relative to the density $\rho_{\text{sub.p}}$ of the particle. Accordingly, selecting bonding paste **332** with a relatively high density can also increase velocity v . By increasing velocity v by increasing applied temperature to bonding paste **332** and/or decreasing viscosity u of bonding paste **332**, particles within bonding paste **332** may experience greater buoyancy, leading to not only an increased likelihood of egress (resulting in greater homogeneity of bonding paste **332**) but also faster egress (resulting in less manufacturing time). It should be noted that air bubbles, flux droplets, impurity particles, etc. are, individually, of lower density as compared to that of bonding paste **332** in order to egress the air bubbles, flux droplets, impurity particles, respectively, to the surface of meniscus **340**. Also, with increasing temperature, thermal expansion of air bubbles may increase radius r and decrease particle density $\rho_{\text{sub.p}}$, thus further compounding the effects that increase their egress velocity v .

(35) Referring to FIG. 4C, thermal module **320** is removed from the heating apparatus, and bonding paste **332** freezes (i.e., the temperature of bonding paste **332** falls below its melting point). The capillary pressure may cause bonding paste **332** to extend between end surface **342** of wall **326b** and wall **346** (e.g., major wall), and into internal volume **335**, particularly during instances of a relatively small gap between walls **326b** and **346**. For example, as shown in the enlarged view, bonding paste **332** may enter internal volume **335** and also form a meniscus **348** (e.g., concave meniscus). However, due to clamps **339a** and **339b** (both shown in FIG. 4A), bonding paste **332** within internal volume **335** may be limited such that bonding paste **332** does not contact a wick structure (e.g., wick structure **328** shown in FIG. 4A). Also, angle **334** between walls **326a** and **326b** can be adjusted to influence capillary rise in bonding paste **332** in molten state so as to regulate movement of bonding paste **332** within internal volume **335**. Additionally, an angle **333** between walls **346** and **326b** is the complement of angle **334**, resulting in angle **333** being larger (e.g., multiple orders of magnitude) than angle **334**. Angle **333** may cause a meniscus **348** with the same radius of curvature and the same contact angle as that of radius of curvature r and contact angle θ , respectively, (shown in FIG. 4B) to form internal to the nearest point between the two parts since the capillary pressure gradient is the same at both menisci. Therefore, the body of bonding paste **332** that forms internally with respect to the nearest point between the two structures (e.g., wall **326b** and wall **346**) is self-limiting and relatively small, reducing the risk that bonding paste **332** can reach an internal wick structure (e.g., wick structure **328** shown in FIG. 4A). Further, droplets **336** and air bubbles **338** (shown in FIGS. 4A and 4B) egress from bonding paste **332**. Beneficially, by removing droplets **336** and air bubbles **338**, bonding paste **332** is a homogenous structure, thus increasing the joint strength and seal hermeticity capabilities of bonding paste **332** after bonding paste **332** has frozen into a solid body.

(36) FIGS. 5-14 show and describe alternate examples of thermal modules. The thermal modules shown and described in FIGS. 5-14 may include any features shown and/or described herein, and may provide modifications to prior examples of thermal modules.

(37) Referring to FIG. 5, a thermal module **420** includes a part **424 a** with a wall **426 a**, as well as a part **424 b** with a wall **426 b**. A bonding paste **432** forms a joint and a hermetic seal between walls **426 a** and **426 b**. As shown in the enlarged view, walls **426 a** and **426 b** are diverging walls and are separated by an angle **434**. As shown, angle **434** is 10 degrees (or approximately 10 degrees). Based on angle **434** being relatively smaller as compared to a prior example, walls **426 a** and **426 b** are closer together. Also, angle **434** between walls **426 a** and **426 b** can be adjusted to influence capillary rise in bonding paste **432** in molten state so as to regulate movement of bonding paste **432** into internal volume **435**. Additionally, an angle **433** between walls **426 a** and **426 b** is the

compliment of angle **434**, resulting in angle **433** being larger than angle **434**. Similar to a prior example, the body of bonding paste **432** that forms internally with respect to the nearest point between the two structures (e.g., part **426 b** and wall **446**) is self-limiting and relatively small, reducing the risk that bonding paste **432** can reach a wick structure **428**. Further, droplets and/or air bubbles (not shown in FIG. 5) egress from bonding paste **432**. Beneficially, by removing droplets and air bubbles, bonding paste **432** is a homogenous structure, thus increasing the joint strength and seal hermeticity capabilities of bonding paste **432** after bonding paste **432** has frozen into a solid body. It should be noted that if the wall thickness of wall **426 b** is not uniform, then angle **433** will not be the exact compliment of angle **434**. However, parts **424 a** and **424 b** may still be designed such that angle **433** is substantially larger than angle **434**. Alternatively, in some examples, both angles **433** and **434** are equal, in which case two nearly identical columns of molten solder (e.g., bonding paste **432**) form similarly in each of the divergent wall clearances (e.g., between walls **426 a** and **426 b**) and thus achieve a similar means of hermetically sealing a structural joint.

(38) Referring to FIG. 6, a part **524b** of a thermal module includes several openings. For example, part **524b** includes a wall **526b** with several openings, such as an opening **552a** and an opening **552b** (representative of several additional openings throughout wall **526b**) formed on an end surface **542** of wall **526b**. Thus, the part **524b** may be referred to as a scalloped part. While a discrete number of openings are shown distributed along the edges of wall **526b**, the number of openings may vary.

(39) Referring to FIG. 7, part **524b** is coupled with a part **524a** to form a thermal module **520**. Part **524a** may include a wall **526a**, with walls **526a** and **526b** forming diverging gap with an angle **534** between walls **526a** and **526b**. End surface **542** of wall **526b** is positioned at a similar angle, as that of angle **534**, with respect to a wall **546** of part **524a**. In one embodiment, bonding paste **532** is applied at the corner of walls **546** and **526a** before installation of inner part **542b**. This results in some portion of the bonding paste **532** being squeezed to the inside of wall **526b** and end surface **542** of wall **526b** during the assembly of parts **524a** and **524b**. Once heated to a molten state, bonding paste **532** may be drawn through openings **552a** and **552b** (as well as other openings in wall **526b**) by wicking forces, or capillary forces, such that at least some of bonding paste **532** passes across wall **526b** and out of internal volume **535** between parts **524a** and **524b**.

(40) Referring to FIG. 8, a thermal module **620** is shown with a part **624a** and a part **624b** having a wall **626a** and a wall **626b**, respectively. As shown, walls **626a** and **626b** may be characterized as diverging walls. Wall **626b** (an interior wall) is generally a vertical wall (extending along a Z-axis) formed at a 90-degree angle with respect to another wall of part **624b**. However, wall **626a** (an exterior wall) is at an angle **634** (e.g., non-zero angle) with respect to wall **626b**. Thus, wall **626a**, representing an external or exterior wall, of part **624a** may provide a wall that causes divergence between walls **626a** and **626b**.

(41) Referring to FIG. 9, a thermal module **720** is shown with a part **724a** and a part **724b** having a wall **726a** and a wall **726b**, respectively. As shown, walls **726a** and **726b** may be characterized as diverging walls. However, walls **726a** and **726b** may include different lengths. For example, wall **726b** is longer than wall **726a**. As a result, wall **726b** may provide a larger internal volume **735** for thermal module **720**, while wall **726a** is shorter to save on materials.

(42) Referring to FIG. 10, a thermal module **820** is shown with a part **824a** and a part **824b** having a wall **826a** and a wall **826b**, respectively. As shown, walls **826a** and **826b** may be characterized as diverging walls. However, wall **826a** may be modified to provide a straight portion **854a** (aligned with wall **826b**) and a divergent portion **854b**. Based on divergent portion **854b** of wall **826b**, wall **826a** and **826b** are diverging walls, thus providing a location for a bonding paste **832**.

(43) Referring to FIG. 11, a thermal module **920** is shown with a part **924a** and a part **924b** having a wall **926a** and a wall **926b**, respectively. The gap formed between walls **926a** and **926b** is divergent due to walls **926a** and **926b** being formed with different angles as described in previous embodiments. A wick structure (not shown) is placed between parts **924a** and **924b** prior to

assembly. As described herein, parts **924a** and **924b** can be combined by using a bonding paste (not shown in FIG. **11**), which requires a heating operation to reflow the bonding paste. Afterward, a refrigerant (not shown in FIG. **11**) is added into thermal module **920** after the heating operation. In order to provide the refrigerant, parts **924a** and **924b** can be modified to include an opening or fill port. For example, wall **926a** may include a port **956** that provides an opening into the internal volume of thermal module **920** to provide a refrigerant. Further, wall **926b** may include an extension **958**, which can be modified (e.g., bent) to cover port **956**, thus acting as a door.

(44) Referring to FIG. **12**, parts **924a** and **924b** may be secured together by a bonding paste in a manner described herein. When a refrigerant **930** (shown as dotted lines) is added to thermal module **920** via port **956**, port **956** can be closed by bending extension **958** to cover port **956** and retain refrigerant **930**, as shown in FIG. **12**. Port **956** can further be sealed by adding bonding paste (not shown in FIG. **12**) to provide a seal between port **956** and extension **958**. Alternatively, in some examples, port **956** can exclude extension **958** and use an alternate method of closure where a wire is partially inserted into port **956** and solder is melted into the gaps between said wire and port **956** to seal port **956** shut for the purpose of retaining the refrigerant.

(45) Referring to FIG. **13**, a thermal module **1020** includes a body may extend in three dimensions. As shown, thermal module **1020** includes a portion **1060a** and a portion **1060b**, and further includes a portion **1060c** representing an elevated or raised portion with respect to portions **1060a** and **1060b**. As shown, portions **1060a** and **1060b** are thermally coupled to a thermal component **1061a** and a thermal component **1061b**, respectively. In one or more implementations, each of thermal component **1061a** and thermal component **1061b** takes the form of a fin stack. Also, **1060c** is thermally coupled to a thermal component **1061c**. In one or more implementations, thermal component **1061c** takes the form of a heat spreader. Based on portion **1060c** being elevated/raised, thermal module **1020** may pass over one or more components of an electronic device, thus allowing thermal module **1020** to conform to a design layout of the electronic device.

(46) Referring to FIG. **14**, a thermal module **1120** includes a body may extend in three dimensions. As shown, thermal module **1020** includes a portion **1160a** and a portion **1160b**, with portions **1160a** and **1160b** representing curved portions. Based on portion **1160a** and **1160b** being curved, thermal module **1120** may pass around one or more a components of an electronic device.

(47) FIGS. **15A** and **15B** represent additional electronic device that may include one or more thermal modules shown and/or described herein. The electronic device shown and described in FIGS. **15A** and **15B** may include one or more heat-generating components, with one or more thermal modules shown and/or described herein that is/are thermally coupled to the one or more heat-generating components. Referring to FIG. **15A**, an electronic device **1200** is shown. Electronic device **1200** may take the form of a mobile wireless communication device, such as a smartphone or a tablet computing device, as non-limiting examples.

(48) As shown, electronic device **1200** includes a housing **1202** and a display **1204** coupled to housing **1202**. Housing **1202** may include metal (including metal alloy), a transparent material (e.g., glass sapphire), or a combination thereof. Display **1204** is designed to present visual information, such as textual information, still images, or motion images (e.g., video). Further, display **1204** may include a capacitive touch sensitive layer, thus allowing display **1204** to receive touch inputs and/or gestures through interaction with display **1204**.

(49) Electronic device **1200** may further include one or more buttons. For example, electronic device **1200** included a button **1206a** and a button **1206b**, each of which can be depressed to provide an input to, for example, interact with and alter/update the visual information on display **1204**. Although buttons **1206a** and **1206b** are shown in particular locations, buttons **1206a** and **1206b** may generally be positioned in other locations. Also, although buttons **1206a** and **1206b** represent a discrete number of buttons, electronic device **1200** may include a different number of buttons.

(50) Referring to FIG. **15B**, an electronic device **1300** takes the form of a desktop computing

device or a standalone display. Electronic device **1300** includes a housing **1302** (e.g., display housing) and a stand **1308**. Housing **1302** and stand **1308** may be rotationally coupled together, thus allowing relative rotational movement between housing **1302** and stand **1308**. As shown, housing **1302** carries a display **1304** designed to present visual information. Although not shown, processing circuitry can be electrically connected to display **1304** using circuits (e.g., flexible circuits).

(51) Although not expressly shown, electronic devices may take the form of various electronic devices. As a non-limiting example, an electronic device may include a video streaming device.

(52) Referring to FIG. **16**, a flowchart **1400** showing a method for forming a thermal module is shown. The flowchart **1400** may be used to form one or more thermal modules shown and/or described herein.

(53) In step **1402**, parts (e.g., housing parts) are provided. The parts may include copper, copper alloy, or another relatively high thermally conductive metal(s). Further, at least one of the parts may include walls that forms a 90-degree, or approximately 90-degree, angle with respect to another wall or surface of the part, while the other part may include walls formed an angle that forms a 60- to 80-degree, or approximately 60- to 80-degree, angle with respect to another wall or surface of the part. As a result, the respective walls of the parts may be diverging walls that form an angle of 10 to 30 degrees, or approximately 10 to 30 degrees.

(54) In step **1404**, a bonding paste is applied to the parts. The bonding paste may include solder paste or copper paste, as non-limiting examples. The bonding paste may be placed between the diverging walls.

(55) In step **1406**, heat is applied to the thermal module. The thermal module may be placed in a brazing furnace, for example, which applies sufficient heat to reflow the bonding paste. Based on the diverging walls, the bonding paste may incur capillary forces that cause the bonding paste to extend further between the diverging walls, thus enhancing the joint and sealing capabilities. Additionally, the bonding paste, in liquid state due to the applied heat, can force out particles, debris, and/or air bubbles, thus providing a homogeneous structure.

(56) In step **1408**, a refrigerant is added. For example, the refrigerant may be added between the parts. The refrigerant may include water, as a non-limiting example. When the thermal module is in use, the refrigerant may be in a liquid state, a gas state, or a combination thereof.

(57) Optionally, in step **1410**, a port (used to add the refrigerant) is sealed. In one or more implementations, one of the first part or the second part includes a port and the other part includes an extension, or door, that can cover the port (after adding the refrigerant). The port can subsequently be shut and sealed by, for example, a bonding paste.

(58) Additionally, prior to step **1402**, a cleaning operation may be performed on the parts. For example, the parts may be placed in an ultrasonic cleaner. Also, an opening or hole may be formed in one of the parts. The opening may provide a vent for pressure equalization between an internal volume of a thermal module and the surrounding air. In particular, when heat is applied in step **1406**, pressure within the thermal module may not increase substantially, based on the opening, and cause unwanted issues (e.g., parts decoupling). The opening can be sealed subsequent to step **1406**.

(59) Referring to FIG. **17**, a flowchart **1500** showing a method for joining housing parts that form an enclosure is shown. The flowchart **1500** may be used to form one or more thermal modules shown and/or described herein.

(60) In step **1502**, a bonding material is provided between i) a first wall of a first part and ii) a second wall of a second part. The bonding material may include a bonding paste that is deposited between the first wall and the second wall. The first wall and the second wall may diverge from each other.

(61) In step **1504**, based on a capillary force, the bonding material is caused to extend between a gap formed by the first wall and the second wall while the bonding material is in a molten or liquid state. The capillary force may be based on the first wall and the second wall being divergent walls.

(62) In step 1506, at an interface of the bonding material and air, a meniscus is formed based on the capillary force. The bonding material may subsequently solidify into a solid body that form a joint between the first part and the second part. The solid body may also form a hermetic seal between the first part and the second part.

(63) Various examples of aspects of the disclosure are described below as clauses for convenience. These are provided as examples, and do not limit the subject technology.

(64) Clause A: A thermal module may include a first part that includes a first wall. The thermal module may further include a second part that includes a second wall. The second wall may be non-parallel with respect to the first wall, and the first wall and the second wall define a gap. The thermal module may further include a bonding paste configured to: extend along the gap between the first wall and the second wall, secure the first wall with the second wall, and provide a hermetic seal between the first wall and the second wall.

(65) Clause B: A thermal module may include a first part that includes a first wall. The first part may define an internal volume. The thermal module may further include a second part disposed in the internal volume. The second part may include a second wall. The thermal module may further include a bonding paste positioned between the first wall and the second wall. In some examples, based on a separation between the first wall and the second wall, the bonding paste forms a meniscus.

(66) Clause C: An electronic device may include a heat-generating component. The electronic device may further include a thermal module thermally coupled to the heat-generating component. The thermal module may include a first part that includes a first wall. The thermal module may further include a second part that includes a second wall. The second wall may be non-parallel with respect to the first wall, and the first wall and the second wall define a gap. The thermal module may further include a bonding paste configured to: extend along the gap between the first wall and the second wall, secure the first wall with the second wall, and provide a hermetic seal between the first wall and the second wall.

(67) Clause D: A method for joining housing parts of a thermal module is described. The method may include providing a bonding paste between i) a first wall of a first part and ii) a second wall of a second part. The first wall and the second wall may diverge from each other. The method may further include applying heat to melt the bonding paste. The method may further include causing, based on a capillary force, the bonding paste to extend between a gap formed by the first wall and the second wall. The capillary force may be based on the first wall and the second wall being divergent walls. The method may further include forming, at the bonding paste, a meniscus based on the capillary force.

(68) Clause E: A method for joining housing parts that form an enclosure is described. The method may include providing a bonding material between i) a first wall of a first part and ii) a second wall of a second part. The bonding material may include a bonding paste. The first wall and the second wall may diverge from each other. The method may further include causing, based on a capillary force, the bonding material to extend between a gap formed by the first wall and the second wall while the bonding material is in a molten or liquid state. The capillary force may be based on the first wall and the second wall being divergent walls. The method may further include forming, at an interface of the bonding material and air, a meniscus based on the capillary force. The bonding material may subsequently solidify into a solid body that form a joint between the first part and the second part. The solid body may also form a hermetic seal between the first part and the second part.

(69) One or more of the above clauses can include one or more of the features described below. It is noted that any of the following clauses may be combined in any combination with each other, and placed into a respective independent clause, e.g., clause A, B, C, D or E.

(70) Clause 1: wherein the first wall is separated from the second wall by an angle of at least 10 degrees.

(71) Clause 2: wherein the first wall is separated from the second wall by an angle of at least 20 degrees.

(72) Clause 3: wherein: the first wall comprises an external wall, and the second wall comprises interior wall that is internal with respect to the first wall.

(73) Clause 4: wherein the first wall includes: an end surface, and openings formed at the end surface.

(74) Clause 5: wherein the bonding paste includes a meniscus.

(75) Clause 6: wherein the bonding paste is selected from solder paste or copper paste.

(76) Clause 7: wherein the hermetic seal forms a joint between the first part and the second part.

(77) Clause 8: wherein: the first part further includes a third wall, and the first wall and the third wall form a right angle.

(78) Clause 9: wherein the first wall is separated from the second wall by an angle of at least 10 degrees.

(79) Clause 10: wherein the first wall is separated from the second wall by an angle of at least 20 degrees.

(80) Clause 11: wherein based on the separation, the bonding paste, in a liquid state, is configured to extend along a gap between the first wall and the second wall.

(81) Clause 12: wherein the bonding paste forms a hermetic seal between the first part and the second part.

(82) Clause 13: wherein the first part further includes: a port configured to receive a liquid, and an extension configured to cover the port and retain the liquid.

(83) Clause 14: wherein: the first part further includes a third wall, the first wall and the third wall form a right angle, and the first wall is separated from the second wall by an angle of at least 10 degrees.

(84) Clause 15: wherein the hermetic seal forms a joint between the first part and the second part.

(85) Clause 16: wherein the bonding paste forms a meniscus.

(86) Clause 17: wherein the thermal module further includes: a wick structure formed on at least one of the first part or the second part.

(87) It is well understood that the use of personally identifiable information should follow privacy policies and practices that are generally recognized as meeting or exceeding industry or governmental requirements for maintaining the privacy of users. In particular, personally identifiable information data should be managed and handled so as to minimize risks of unintentional or unauthorized access or use, and the nature of authorized use should be clearly indicated to users.

(88) As used herein, the phrase “at least one of” preceding a series of items, with the term “and” or “or” to separate any of the items, modifies the list as a whole, rather than each member of the list (i.e., each item). The phrase “at least one of” does not require selection of at least one of each item listed; rather, the phrase allows a meaning that includes at least one of any one of the items, and/or at least one of any combination of the items, and/or at least one of each of the items. By way of example, the phrases “at least one of A, B, and C” or “at least one of A, B, or C” each refer to only A, only B, or only C; any combination of A, B, and C; and/or at least one of each of A, B, and C.

(89) The predicate words “configured to”, “operable to”, and “programmed to” do not imply any particular tangible or intangible modification of a subject, but, rather, are intended to be used interchangeably. In one or more implementations, a processor configured to monitor and control an operation or a component may also mean the processor being programmed to monitor and control the operation or the processor being operable to monitor and control the operation. Likewise, a processor configured to execute code can be construed as a processor programmed to execute code or operable to execute code.

(90) When an element is referred to herein as being “connected” or “coupled” to another element, it is to be understood that the elements can be directly connected to the other element, or have

intervening elements present between the elements. In contrast, when an element is referred to as being “directly connected” or “directly coupled” to another element, it should be understood that no intervening elements are present in the “direct” connection between the elements. However, the existence of a direct connection does not exclude other connections, in which intervening elements may be present.

(91) Phrases such as an aspect, the aspect, another aspect, some aspects, one or more aspects, an implementation, the implementation, another implementation, some implementations, one or more implementations, an embodiment, the embodiment, another embodiment, some embodiments, one or more embodiments, a configuration, the configuration, another configuration, some configurations, one or more configurations, the subject technology, the disclosure, the present disclosure, other variations thereof and alike are for convenience and do not imply that a disclosure relating to such phrase(s) is essential to the subject technology or that such disclosure applies to all configurations of the subject technology. A disclosure relating to such phrase(s) may apply to all configurations, or one or more configurations. A disclosure relating to such phrase(s) may provide one or more examples. A phrase such as an aspect or some aspects may refer to one or more aspects and vice versa, and this applies similarly to other foregoing phrases.

(92) The word “exemplary” is used herein to mean “serving as an example, instance, or illustration”. Any embodiment described herein as “exemplary” or as an “example” is not necessarily to be construed as preferred or advantageous over other embodiments. Furthermore, to the extent that the term “include”, “have”, or the like is used in the description or the claims, such term is intended to be inclusive in a manner similar to the term “comprise” as “comprise” is interpreted when employed as a transitional word in a claim.

(93) All structural and functional equivalents to the elements of the various aspects described throughout this disclosure that are known or later come to be known to those of ordinary skill in the art are expressly incorporated herein by reference and are intended to be encompassed by the claims. Moreover, nothing disclosed herein is intended to be dedicated to the public regardless of whether such disclosure is explicitly recited in the claims. No claim element is to be construed under the provisions of 35 U.S.C. § 112, sixth paragraph, unless the element is expressly recited using the phrase “means for” or, in the case of a method claim, the element is recited using the phrase “step for”.

(94) The previous description is provided to enable any person skilled in the art to practice the various aspects described herein. Various modifications to these aspects will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other aspects. Thus, the claims are not intended to be limited to the aspects shown herein, but are to be accorded the full scope consistent with the language claims, wherein reference to an element in the singular is not intended to mean “one and only one” unless specifically so stated, but rather “one or more”. Unless specifically stated otherwise, the term “some” refers to one or more. Pronouns in the masculine (e.g., his) include the feminine and neuter gender (e.g., her and its) and vice versa. Headings and subheadings, if any, are used for convenience only and do not limit the subject disclosure.

Claims

1. A thermal module, comprising: a first part comprising a first wall; a second part comprising a second wall, wherein the second wall is angled such that the second wall is non-parallel with respect to the first wall to define a diverging portion, and the first wall and the second wall define a gap based on the diverging portion; and a bonding paste configured to: extend along and fill the gap between the first wall and the second wall, secure and engage the first wall with the second wall at the diverging portion, and provide a hermetic seal between the first wall and the second wall.

2. The thermal module of claim 1, wherein the first wall is separated from the second wall by an angle of at least 10 degrees.
 3. The thermal module of claim 1, wherein the first wall is separated from the second wall by an angle of at least 20 degrees.
 4. The thermal module of claim 1, wherein: the first wall comprises an external wall, and the second wall comprises interior wall that is internal with respect to the first wall.
 5. The thermal module of claim 4, wherein the first wall comprises: an end surface, and openings formed at the end surface.
 6. The thermal module of claim 1, wherein the bonding paste comprises a meniscus.
 7. The thermal module of claim 1, wherein the bonding paste is selected from solder paste or copper paste.
 8. The thermal module of claim 1, wherein the hermetic seal forms a joint between the first part and the second part.
 9. A thermal module, comprising: a first part comprising a first wall, wherein the first part defines an internal volume; a second part disposed in the internal volume, the second part comprising a second wall; and a bonding paste positioned between the first wall and the second wall, wherein based on a separation between the first wall and the second wall, the bonding paste forms a concave meniscus.
 10. The thermal module of claim 9, wherein: the first part further comprises a third wall, and the first wall and the third wall form a right angle.
 11. The thermal module of claim 10, wherein the first wall is separated from the second wall by an angle of at least 10 degrees.
 12. The thermal module of claim 10, wherein the first wall is separated from the second wall by an angle of at least 20 degrees.
 13. The thermal module of claim 9, wherein based on the separation, the bonding paste, in a liquid state, is configured to extend along a gap between the first wall and the second wall.
 14. The thermal module of claim 9, wherein the bonding paste forms a hermetic seal between the first part and the second part.
 15. The thermal module of claim 9, wherein: concave meniscus is formed in part based on a pressure on the bonding paste based on an angle between the first part and the second part, and the angle is based on the separation.
 16. An electronic device, comprising: a heat-generating component; and a thermal module thermally coupled to the heat-generating component, the thermal module comprising: a first part comprising a first wall and a second wall; a second part comprising a third wall, wherein the third wall is non-parallel with respect to the first wall and the second wall, and the first wall and the third wall define a gap; and a bonding paste engaged with the first wall, the second wall, and the third wall, the bonding paste configured to: extend along the gap between the first wall and the second wall, secure the first wall with the second wall, form a first concave meniscus between the first wall and the third wall, form a second concave meniscus between the second wall and the third wall, and provide a hermetic seal between the first wall and the second wall.
 17. The electronic device of claim 16, wherein: the first wall and the third wall form a right angle, and the first wall is separated from the second wall by an angle of at least 10 degrees.
 18. The electronic device of claim 16, wherein the hermetic seal forms a joint between the first part and the second part.
 19. The electronic device of claim 16, wherein the bonding paste forms a meniscus.
 20. The electronic device of claim 16, wherein the thermal module further comprises a wick structure formed on the first part or the second part.
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